3. It aims to enable the poor people to meet their water needs without depending on subsidies.

Which of the statements given above are correct ?

(a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

Ans. (c): Statement 1 is correct. Water Credit is a powerful solution and the first to put micro finance tools to work in the water and sanitation sector. Water Credit helps bring small loans to those who need access to affordable financing and expert resources to make household water and toilet solutions a reality.

Statement 2 is incorrect. Water credit is an initiative of Water.org. Water.org is a global non-profit organization working to bring water and sanitation to the world.

Statement 3 is correct. It helps people get access to safe water and sanitation through affordable financing, such as small loans.

58. In India, the central bank's function as the 'lender of last resort' usually refers to which of the following?

1. Lending to trade and industry bodies when they fail to borrow from other sources
2. Providing liquidity to the banks having a temporary crisis
3. Lending to governments to finance budgetary deficits

Select the correct answer using the code given below.

(a) 1 and 2 (b) 2 only
(c) 2 and 3 (d) 3 only

Ans. (b): In India, the central Bank's function as the 'lender of last resort' usually refers to providing liquidity to the Banks having a temporary crisis.
Hence, option (b) is the correct answer.

SCIENCE

Physics

59. In a pressure cooker, the temperature at which the food is cooked depends mainly upon which of the following?

1. Area of the hole in the lid
2. Temperature of the flame
3. Weight of the lid

Which of the statements given above is/are correct?

(a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

Ans. (c): Inside a pressure cooker the boiling point is increased by increasing the pressure. Hence cooking is faster. Therefore, the temperature at which food is cooked is affected by the pressure and not temperature of the flame. This can also be understood by understanding why cooking is difficult on hills. At high altitudes, atmospheric pressure is lower, reducing the boiling point of water as compared to that at sea level.

60. Water can dissolve more substances than any other liquid because

(a) it is dipolar in nature
(b) it is a good conductor of heat
(c) it has high value of specific heat
(d) it is an oxide of hydrogen

Ans. (a): Water is called the Universal Solvent it is capable of dissolving a variety of different substances more than any other liquid. Water can dissolve more substances than any other compound because of its polar nature.
Hence, option (a) is the correct answer.

Chemistry

61. Bisphenol A (BPA), a cause of concern, is a structural/key component in the manufacture of which of the following kinds of plastics?

(a) Low-density polyethylene
(b) Polycarbonate
(c) Polyethylene terephthalate
(d) Polyvinyl chloride

Ans. (b): Bisphenol A (BPA) is an industrial chemical used to make polycarbonate, a hard, clear plastic, which is used in many consumer products. BPA is also found in epoxy resins, which act as a protective lining on the inside of some metal-based food and beverage cans. Bisphenol A can leach into food from the protective internal epoxy resin coatings of canned foods and from consumer products such as polycarbonate tableware, food storage containers, water bottles, and baby bottles. The degree to which BPA leaches from polycarbonate bottles into liquid may depend more on the temperature of the liquid or bottle, than the age of the container. BPA can also be found in breast milk. One reason people may be concerned about BPA is because human exposure to BPA is widespread.
Hence, option (b) is the correct answer.

Zoology

62. Consider the following statements :

1. Adenoviruses have single-stranded DNA genomes whereas retroviruses have double-stranded DNA genomes.
2. Common cold is sometime caused by an adenovirus whereas AIDS is caused by a retrovirus.

Which of the statements given above is/are correct?

(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

Ans. (b): It is actually the opposite the statement given. Adenoviruses have double stranded DNA genomes while Retroviruses have single stranded DNA genomes. So, statement 1 is not correct.
Common cold can sometimes be caused by adenoviruses while AIDS is caused by retrovirus. So, statement 2 is correct. Therefore, the correct answer is (b).

Botany

63. Consider the following :

1. Bacteria
2. Fungi
3. Virus

Which of the above can be cultured in artificial/synthetic medium?

(a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

Ans. (a): Options 1 and 2 are correct:
Microbes like bacteria and many fungi can be grown on nutritive media to form colonies that can be seen with the naked eyes. Such cultures are useful in studies on micro -organisms. Option 3 is not correct: Unlike bacteria, many of which can be grown on an artificial nutrient medium, viruses require a living host cell for replication. Infected host cells (eukaryotic or prokaryotic) can be cultured and grown, and then the growth medium can be harvested as a source of virus.
Hence, option (a) is the correct answer.

Science & Bio- technology

64. With reference to recent developments regarding 'Recombinant Vector Vaccines', consider the following statements :

1. Genetic engineering is applied in the development of these vaccines.
2. Bacteria and viruses are used as vectors.

Which of the statements given above is/are correct?

(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

Ans. (c): As per the given statements in reference to recent developments regarding 'Recombinant Vector Vaccines' both the given statements are correct.
The recombinant virus vectors vaccine is one of the important emerging technology are live replicating viruses that are engineered to carry extra genes derived from a pathogen and these extra genes produce proteins against which we want to generate immunity.
It exposes the body to proteins made by a virus as bacteria, are often made by a using weakened or inactive versions of that virus or bacteria.
Hence, option (c) is the correct answer.

65. In the context of hereditary diseases, consider the following statements:

1. Passing on mitochondrial diseases from parent to child can be prevented by mitochondrial replacement therapy either before or after in vitro fertilization of egg.
2. A child inherits mitochondrial diseases entirely from mother and not from father.

Which of the statements given above is/are correct?

(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

Ans. (c): Mitochondrial Replacement Therapy(MRT), sometimes called mitochondrial donation, is the replacement of mitochondria in one or more cells to prevent or ameliorate disease.
MRT originated as a special form of In Vitro Fertilisation in which some or all of the future baby's mitochondrial DNA (mtDNA) comes from a third party. This technique is used in cases when mothers carry genes for mitochondrial diseases.
In Pronuclear transfer technique MRT is done after In vitro fertilisation (IVF). Hence Statement 1 is true.
A child inherits mitochondrial diseases only from Mother. Hence, Statement 2 is true.
So, option (c) is the correct answer.

66. Bollgard I and Bollgard II technologies are mentioned in the context of:

(a) clonal propagation of crop plants
(b) developing genetically modified crop plants
(c) production of plant growth substances
(d) production of biofertilizers

Ans. (b) : Bollgard I, the predominantly cultivated pest tolerant cotton, contains only one gene, the Cry 1Ac, from Bacillus thuringiensis. Bollgard II contains the Cry 2 Ab gene, in addition to Cry 1 Ac.
Hence, option (b) is the correct answer.

67. With reference to street-lighting, how do sodium lamps differ from LED lamps?

1. Sodium lamps produce light in 360 degrees but it is not so in the case of LED lamps.
2. As street-lights, sodium lamps have longer life span that LED lamps.
3. The spectrum of visible light from sodium lamps is almost monochromatic while LED lamps offer significant colour advantages in street-lighting.

Select the correct answer using the code given below.

(a) 3 only (b) 2 only
(c) 1 and 3 only (d) 1, 2 and 3

Ans. (c): Sodium lamps are multi-directional that they emit light in 360 degrees, while led lamps emit light in 180 degrees. Hence this statement 1 is correct.
Lifespan of LED is more than Sodium lamps. Hence, this statement is incorrect.
Sodium lamps are monochromatic and LEDs can be designed to generate the entire spectrum of visible light colors without having to use the traditional color filters required by traditional lighting solutions. This is correct statement.

68. 'Triclosan', considered harmful when exposed to high levels for a long time, is most likely present in which of the following?

(a) Food preservatives
(b) Fruit-ripening substances
(c) Reused plastic containers
(d) Toiletries

Ans. (d): Triclosan is a lipophilic compound with antimicrobial properties. It is an ingredient added to many consumer products intended to reduce or prevent bacterial contamination. It is added to some antibacterial soaps and body washes, toothpastes, and some cosmetics. It also can be found in clothing, kitchenware, furniture, and toys.
Hence, option (d) is the correct answer.

69. Which one of the following is a reason why astronomical distances are measured in light-years?